Ej 2

$\forall t, u :: AB\ a$ $\quad$ altura $t \geq$ altura (zipAB t u)

$\forall t :: AB\ a \cdot P(t)$ $\quad$ con

$P(t) \equiv \forall u :: AB\ a$ $\quad$ altura $t \geq$ altura (zipAB t u)

Por inducción estr. en AB a, si

- $P(Nil)$
- $\forall i, d :: AB\ a\ r\ a \cdot \underbrace{P(i) \wedge P(d)}_{HI} \Rightarrow P(Bin\ i\ r\ d)$

entonces sabemos $\forall t :: AB\ a \cdot P(t)$

Plan

▷ Caso $P(Nil)$

· ecuaciones directas para un u cualquiera

▷ Caso $P(Bin\ i\ r\ d)$

Sean $i\ r\ d$ cualesquiera

asumimos $P(i) \wedge P(d)$

Qvq $P(Bin\ i\ r\ d)$

↳ $\forall u : AB\ a \ \ldots$

$u = Nil$

LEMA { ✓

o

$u = Bin\ u_i\ u_r\ u_d$

{ ✓

× lema de gen en AB

Lema ℤ

1. $\max a\ b = \max b\ a$
2. $b \geq c \Rightarrow \max a\ b \geq \max a\ c$
3. $b \geq c \Rightarrow \max b\ a \geq \max c\ a$

▷ Caso $P(Nil)$

Qvq $\forall u::AB\alpha \cdot altura\ (Nil) \geq altura\ (zip\ AB\ Nil\ u)$

Sea $u::AB\alpha$ cualquiera

Qvq

$altura\ (Nil) \geq altura\ (zip\ AB\ Nil\ u)$

$\equiv \{Z_0\}$ $altura\ (Nil) \geq altura\ (const\ Nil\ u)$

$\equiv \{C\}$ $altura\ Nil \geq altura\ ((\backslash x \to \backslash y \to x)\ Nil\ u)$

$\equiv \{B\}$ $altura\ Nil \geq altura\ ((\backslash y \to Nil)\ u)$

$\equiv \{B\}$ $altura\ Nil \geq altura\ Nil$ ☑ (por $x = x \Rightarrow x \geq x$ $\forall x::Int$)

(opcional)

$\equiv \{A\emptyset\}$ $0 \geq altura\ Nil$

$\equiv \{A\emptyset\}$ $0 \geq 0$

☑ por $\mathbb{Z}$

D Caso $P(Bin\ i\ r\ d)$

Sabemos $\underbrace{P(i) \wedge P(d)}_{HI}$

qvq $P(Bin\ i\ r\ d)$

$P(i) \equiv \forall u :: AB\ a \cdot altura\ i \geq altura\ (zipAB\ i\ u)$

$P(d) \equiv \forall u :: AB\ a \cdot altura\ d \geq altura\ (zipAB\ d\ u)$

(pueden ser distintos u

Qvq $P(Bin\ i\ r\ d)$

Qvq $\forall u :: AB\ a$ $\quad altura\ (Bin\ i\ r\ d) \geq altura\ ((zipAB\ (Bin\ i\ r\ d)\ u)$

Sea $u :: AB\ a$ cualquiera

Qvq $altura\ (Bin\ i\ r\ d) \geq altura\ ((zipAB\ (Bin\ i\ r\ d)\ u)$

$\underset{\{AL\}}{=} 1 + \max\ (altura\ i)\ (altura\ d) \geq altura\ (zipAB\ (Bin\ i\ r\ d)\ u)$

por lema de generación en $AB\ a$. $u = Nil$ o

$u = Bin\ u_i\ u_r\ u_d$

para algún $u_i, u_d :: AB\ a$ $\quad u_r :: a$

▷ Caso $u = \text{Nil}$

... pendiente ...

▷ Caso $u = \text{Bin}\ u_i\ u_r\ u_d$

$$\underbrace{1 + \max\ (\text{altura}\ i)\ (\text{altura}\ d)}_{E1} \geq \text{altura}\ (\text{zipAB}\ (\text{Bin}\ i\ r\ d)\ u)$$

qvq

$$E1 \geq \text{altura}\ (\text{zipAB}\ (\text{Bin}\ i\ r\ d)\ (\text{Bin}\ u_i\ u_r\ u_d))$$

$$\overset{\{Z_1\}}{\equiv} \quad E1 \geq \text{altura}\ ((\backslash t \to \text{case}\ t\ d \ldots)\ (\text{Bin}\ u_i\ u_r\ u_d))$$

$$\overset{\{B\}\ \text{case}}{\equiv} \quad E1 \geq \text{altura}\ (\text{Bin}\ (\text{zipAB}\ i\ u_i)\ (r, u_r)\ (\text{zipAB}\ d\ u_d))$$

$$\overset{\{A1\}}{\equiv} \quad E1 \geq 1 + \max\ (\text{altura}\ (\text{zipAB}\ i\ u_i))\ (\text{altura}\ (\text{zipAB}\ d\ u_d))$$

$$\equiv \quad \max\ (\text{altura}\ i)\ (\text{altura}\ d) \geq \max\ (\text{altura}\ (\text{zipAB}\ i\ u_i))\ (\text{altura}\ (\text{zipAB}\ d\ u_d))$$

$$\boxed{\{Z\}}$$

qvq

$$\max\ (\text{altura } i)\ (\text{altura } d) \geq \max\ (\text{altura } (\text{zipAB } i\ u_i))\ (\text{altura } (\text{zipAB } d\ u_d))$$

Sabemos

$P(i) \equiv \forall u :: AB\ a \cdot \text{altura } i \geq \text{altura } (\text{zipAB } i\ u)$

$P(d) \equiv (\forall u :: AB\ a \cdot \text{altura } d \geq \text{altura } (\text{zipAB } d\ u)$

(pueden ser distintos u

① $\max\ a\ b = \max\ b\ a$

② $b \geq c \Rightarrow \max\ ab \geq \max\ a\ c$

③ $b \geq c \Rightarrow \max\ b\ a \geq \max\ c\ a$

Entonces

Por $P(i)$ y ③ en $u = u_i$

$\max\ (\text{altura } i)\ (\text{altura } d) \geq \max\ (\text{altura } (\text{zipAB } i\ u_i))\ (\text{altura } d)$ E2

Por $P(d)$ y ② en $u = u_d$

E2 $\geq \max\ (\text{altura } (\text{zipAB } i\ u_i))\ (\text{altura } (\text{zipAB } d\ u))$ ✓

Ej 3

$$
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\dfrac{}{\Gamma \vdash \sigma \vee (\rho \Rightarrow \tau)}\,\mathrm{Ax}
\qquad
\dfrac{\dfrac{}{\Gamma, \sigma \vdash \sigma}\,\mathrm{Ax}}{\Gamma, \sigma \vdash \sigma \vee \tau}\,\vee i_1
\qquad
\dfrac{\dfrac{\dfrac{}{\Gamma, \rho \Rightarrow \tau \vdash \rho \Rightarrow \tau}\,\mathrm{Ax} \qquad \dfrac{}{\Gamma, \rho \Rightarrow \tau \vdash \rho}\,\mathrm{Ax}}{\Gamma, \rho \Rightarrow \tau \vdash \tau}\,\Rightarrow e}{\Gamma, \rho \Rightarrow \tau \vdash \sigma \vee \tau}\,\vee i_2
}{\Gamma, \rho, (\sigma \vee (\rho \Rightarrow \tau)) \vdash \sigma \vee \tau}\,\vee e
}{\rho \vdash (\sigma \vee (\rho \Rightarrow \tau)) \Rightarrow (\sigma \vee \tau)}\,\Rightarrow i
}{\vdash \rho \Rightarrow ((\sigma \vee (\rho \Rightarrow \tau)) \Rightarrow (\sigma \vee \tau))}\,\Rightarrow i
$$

Ej 4

$\tau ::= \cdots \mid \mathrm{Cola}\ \tau$

$M ::= \cdots \mid \langle\rangle_\tau \mid M \bullet M \mid \mathrm{próx}(M) \mid \mathrm{des}(M)$

$\mid \mathrm{case}\ M\ \mathrm{of}\ \langle\rangle \rightsquigarrow M;\ c \bullet x \rightsquigarrow M$

$$\frac{}{\Gamma \vdash \langle\rangle_\tau : \mathrm{Cola}\ \tau}\ \text{T-VACÍA}$$

$$\frac{\Gamma \vdash M : \mathrm{Cola}\ \tau \quad \Gamma \vdash N : \tau}{\Gamma \vdash M \bullet N : \mathrm{Cola}\ \tau}\ \text{T-AGR.}$$

$$\frac{\Gamma \vdash M : \mathrm{Cola}\ \tau}{\Gamma \vdash \mathrm{próx}(M) : \tau}\ \text{T-próx}$$

$$\frac{\Gamma \vdash M : \mathrm{Cola}\ \tau}{\Gamma \vdash \mathrm{des}(M) : \mathrm{Cola}\ \tau}\ \text{T-DES}$$

$$\frac{\Gamma \vdash M : \mathrm{Cola}\ \tau \quad \Gamma \vdash N : \gamma \quad \Gamma, c : \mathrm{Cola}\ \tau, x : \tau \vdash O : \gamma}{\Gamma \vdash \mathrm{case}\ M\ \mathrm{of}\ \langle\rangle \rightsquigarrow N;\ c \bullet x \rightsquigarrow O : \gamma}\ \text{T-CASE}$$

$V ::= \cdots \mid \langle\rangle_\tau \mid V \bullet V$

$$V ::= \dots \mid \langle\rangle_\tau \mid V \cdot V$$

Reglas de congruencia

$$\frac{M \to M'}{M \cdot N \to M' \cdot N}\ \text{E-IZQ}\cdot \qquad \frac{N \to N'}{V \cdot N \to V \cdot N'}\ \text{E-DER}\cdot$$

$$\frac{M \to M'}{\text{próx}(M) \to \text{próx}(M')}\ \text{E-PROX} \qquad \frac{M \to M'}{\text{des}(M) \to \text{des}(M')}\ \text{E-DES}$$

$$\frac{M \to M'}{\text{case } M \text{ of } \langle\rangle \rightsquigarrow N;\ c \cdot x \rightsquigarrow O \to \text{case } M' \text{ of } \langle\rangle \rightsquigarrow N;\ c \cdot x \rightsquigarrow O}\ \text{E-CASE}$$

Reglas de cómputo

$$\frac{}{\text{próx}(\langle\rangle_\tau \cdot V_1) \to V_1}\ \text{E-PROXV} \qquad \frac{}{\text{próx}((V_1 \cdot V_2) \cdot V_3) \to \text{próx}(V_1 \cdot V_2)}\ \text{E-PROX2}$$

$$\frac{}{\text{des}(\langle\rangle_\tau \cdot V_1) \to \langle\rangle_\tau}\ \text{E-DESV} \qquad \frac{}{\text{des}((V_1 \cdot V_2) \cdot V_3) \to \text{des}(V_1 \cdot V_2) \cdot V_3}\ \text{E-DES2}$$

$$\frac{}{\text{case } \langle\rangle_\tau \text{ of } \langle\rangle \rightsquigarrow M;\ c \cdot x \rightsquigarrow N \longrightarrow M}\ \text{E-CASEV}$$

$$\frac{}{\boxed{\text{case } V_1 \cdot V_2 \text{ of } \langle\rangle \rightsquigarrow M;\ c \cdot x \rightsquigarrow N \rightarrow N\{c := V_1, x := V_2\}}}\ \boxed{\text{E-CASE2}}$$

$$\boxed{\text{case } \underbrace{(\langle\rangle_{Nat} \cdot \underline{1}) \cdot \underline{0}} \text{ of } \langle\rangle \rightsquigarrow \underbrace{\text{próximo}(\langle\rangle_{Bool})};\ c \cdot x \rightsquigarrow \underbrace{\text{iszero}(x)}}$$

$$\xrightarrow[\text{E-CASE2}]{} \boxed{\text{iszero}(x)\{x := \underline{0}\} = \text{iszero}(\underline{0}) \xrightarrow[\text{EISZERO}_0]{} \text{True}}$$

$$\{c := \langle\rangle_{Nat} \cdot \underline{1}\}$$

$$\boxed{\text{último}_\tau := \lambda c : \text{Cola}\,\tau.\ \text{case } c \text{ of } \langle\rangle \rightsquigarrow \text{próximo}(\langle\rangle_\tau);\ k \cdot x \rightsquigarrow x}$$

$$\boxed{\vdash \text{último}_\tau : \text{Cola}\,\tau \to \tau}$$

$$\boxed{M ::= \ldots \mid \bot_\sigma}$$

$$\frac{}{\boxed{\Gamma \vdash \bot_\sigma : \sigma}}\ \text{T-BOTTOM}$$